**WEST BENGAL CORPORATE INTELLIGENCE**

**An Excel-Based Analysis of Company Registrations, LLPs, Industry Distribution and Corporate Status**

**Project Category:** Data Analytics
**Tool:** Microsoft Excel 2021
**Data Source:** MCA Corporate Data Management Portal
**Geographical Scope:** West Bengal, India
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**Year:** 2026

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## 1. Executive Summary

West Bengal Corporate Intelligence is an Excel-based Data Analytics project developed to understand the corporate landscape of West Bengal through company registrations, Limited Liability Partnerships (LLPs), corporate status, industry distribution, geographical distribution and capital information.

The project uses corporate master data obtained from the Ministry of Corporate Affairs (MCA) Corporate Data Management Portal.

Three datasets were collected and consolidated into a structured analytical workbook:

- Company Master Data
- LLP Master Data
- Foreign Company Master Data

The project follows a complete data analytics workflow, beginning with raw data collection and data cleaning, followed by transformation, analysis, PivotTables, PivotCharts and an interactive dashboard.

The final dashboard provides a consolidated visual representation of the analysis through KPI cards, charts, slicers and a geographical map.

The primary objective was not simply to organize corporate records, but to transform large and complex datasets into meaningful information that can support business understanding and data-driven decisions.

## 2. Problem Statement

Corporate master data contains valuable information about registered businesses, but raw datasets are often difficult to interpret directly.

The original datasets contained thousands of records, multiple classifications, inconsistent geographical names, formatting issues and information spread across different files.

Without proper cleaning and analysis, it becomes difficult to answer questions such as:

- How many companies and LLPs are registered in West Bengal?
- Which company classifications are most common?
- Which industries have the highest number of registrations?
- How are businesses distributed across districts?
- What are the different corporate operating statuses?
- How have company and LLP registrations changed over time?
- What do authorized and paid-up capital values indicate?
- What is the foreign-company presence represented in the available MCA dataset?

This project addresses these questions by building a structured analytical system in Microsoft Excel.

**3. Project Objectives**

The major objectives were:

1. Collect corporate registration data from the MCA portal.
2. Organize multiple datasets into a single analytical workbook.
3. Clean and standardize inconsistent data.
4. Identify missing values, duplicates and data-quality issues.
5. Analyze company, LLP and foreign-company records separately.
6. Understand industry-wise and district-wise business distribution.
7. Examine corporate status and registration trends.
8. Analyze authorized and paid-up capital.
9. Build PivotTables and PivotCharts for analytical reporting.
10. Develop an interactive dashboard with KPI cards and slicers.
11. Present findings through a professional Excel-based reporting system.

## 4. Data Collection and Dataset Overview

The data was collected from the Ministry of Corporate Affairs Corporate Data Management Portal.

Three separate datasets were used.

| Dataset | Original Records |
|---|---|
| Company Master | 263,295 |
| LLP Master | 33,932 |
| Foreign Company Master | 5,385 |
| **Total Source Records** | **302,612** |

**Important geographical consideration:**

The Company and LLP datasets were West Bengal-specific. The Foreign Company Master dataset contained India-wide records. Therefore, all 5,385 foreign-company records must not be interpreted as West Bengal registrations. Only records with verified West Bengal addresses can be used to identify West Bengal foreign-company presence.

## 5. Tools and Technologies Used

**Microsoft Excel 2021**

Used as the primary analytical platform for data preparation, calculations, reporting and dashboard development.

**Power Query**

Used for:

- Data transformation
- Removing unnecessary blank records
- Removing duplicate records
- Handling data-format inconsistencies
- Cleaning and preparing structured datasets

**Excel Formulas**

Used for:

- Lookup operations
- Conditional calculations
- Counting and aggregation

- Date-based analysis
- Text cleaning
- Data validation
- Ranking and statistical analysis

**Important functions included:**

XLOOKUP, INDEX, MATCH, VLOOKUP, IF, AND, OR, COUNTIF, COUNTIFS, SUMIF, SUMIFS, AVERAGEIF, MINIFS, MAX, SUMPRODUCT, YEAR, DATE, TRIM, CLEAN, LEFT, RIGHT, MID, FILTER, SORT and UNIQUE.

**PivotTables and PivotCharts**

Used to summarize large datasets, compare categories and generate analytical visualizations.

**Interactive Slicers**

Used to filter PivotTables and connected dashboard charts dynamically.

**6. Workbook Architecture**

The project was organized into separate worksheets to maintain a structured workflow.

| **Sheet** | **Purpose** |
|---|---|
| Project_Overview | Project introduction and scope |
| Raw_Company | Original company dataset |
| Raw_LLP | Original LLP dataset |
| Raw_Foreign | Original foreign-company dataset |
| Clean_Company | Cleaned company data |
| Clean_LLP | Cleaned LLP data |
| Clean_Foreign | Cleaned foreign-company data |
| Data_Quality | Data-quality checks |
| Formula_Library | Reference for Excel formulas |
| Company_Formula_Applied | Company formula-based analysis |
| LLP_Formula_Applied | LLP formula-based analysis |
| Foreign_Formula_Applied | Foreign-company formula-based analysis |
| Combined_Summary_Analysis | Consolidated analytical findings |

| | |
|---|---|
| Pivot_Table | PivotTable calculations and helper summaries |
| Dashboard | Final interactive dashboard |

The workbook maintains separation between original data, cleaned data, calculations, analysis and visualization.

## 7. Data Cleaning and Preparation

Data cleaning was one of the most important stages of this project.

The original MCA datasets required preparation before meaningful analysis could be performed.

The following activities were carried out:

### 7.1 Removal of Blank Records

Unnecessary blank rows were identified and removed to prevent incorrect calculations and PivotTable summaries.

### 7.2 Duplicate Identification and Removal

Duplicate records were examined and removed where appropriate to improve data consistency.

### 7.3 Date Standardization

Registration-date fields were checked and prepared for year-wise analysis.

### 7.4 Capital Data Preparation

Authorized Capital and Paid-up Capital fields were prepared for numerical analysis, including total, average, minimum and maximum calculations.

### 7.5 Geographical Data Standardization

District names were examined to identify inconsistent spellings and alternative naming conventions.

Examples included:

- Calcutta and Kolkata
- Darjiling and Darjeeling
- Koch Bihar and Cooch Behar
- Puruliya and Purulia
- Parganas North and North 24 Parganas
- Parganas South and South 24 Parganas
- East Midnapore and Purba Medinipur

The purpose was to improve geographical consistency and make district-wise analysis more reliable.

### 7.6 Data Quality Checks

The Data_Quality worksheet was designed to monitor issues such as:

- Missing values
- Duplicate records
- Invalid or inconsistent formats
- Data errors
- Incomplete geographical information

## 8. Challenges Encountered and Solutions Implemented

This project involved several practical challenges during data preparation and analysis.

### Challenge 1: Date Values Appearing as Null in Power Query

**Problem:**

Registration dates appeared correctly in the normal Excel worksheet but returned null values in Power Query.

**Solution:**

The date-column data type and transformation process were investigated. Date formatting and type-conversion issues were addressed before using the fields for year-wise analysis.

**Outcome:**

The registration-date fields could be prepared for chronological analysis.

### Challenge 2: Power Query Errors

**Problem:**

During data transformation, Power Query displayed errors, including a reported set of 108 errors.

**Solution:**

The affected data and column types were examined. Inconsistent values and conversion issues were investigated, and cleaning operations were applied.

**Outcome:**

The data preparation process became more structured and reliable.

### Challenge 3: PIN Code Formatting

**Problem:**

PIN codes did not always retain consistent formatting.

**Solution:**

The PIN code column was reviewed and its formatting was standardized to avoid unintended numeric-format changes.

**Outcome:**

Geographical information became easier to inspect and use.

**Challenge 4: Inconsistent District Names**

**Problem:**

The same district could appear under different spellings or alternative names.

**Solution:**

Alternative district names were identified and a standardization approach was developed.

Ambiguous records were kept separate rather than being assigned to districts without sufficient evidence.

**Outcome:**

District-wise reporting became more consistent while reducing incorrect geographical assignments.

**Challenge 5: Foreign Company Geographical Scope**

**Problem:**

The Foreign Company Master dataset contained India-wide records, unlike the West Bengal-specific Company and LLP datasets.

**Solution:**

The foreign-company dataset was treated separately. Only address-verified West Bengal records should be considered for West Bengal-specific foreign-company reporting.

**Outcome:**

The project avoids incorrectly presenting the entire India-wide foreign-company dataset as West Bengal data.

**Challenge 6: Building Registration Trends**

**Problem:**

Registration trends required consistent date values and year-based calculations.

**Solution:**

Date-based COUNTIFS calculations were used to count records within individual calendar years.

**Outcome:**

Year-wise registration analysis was developed for the Company and LLP datasets.

The Foreign Company Master dataset did not contain a registration-date field, so a comparable registration-trend analysis was not possible for that dataset.

**Challenge 7: Dynamic KPI Cards**

**Problem:**

Initially, KPI shapes were displaying fixed values or becoming blank when PivotTables changed their size after slicer selections.

**Solution:**

A helper PivotTable was created using Company Class and Count of Company Class. The relevant slicer was connected to this helper PivotTable.

GETPIVOTDATA was then used to retrieve the filtered Grand Total through a stable helper cell, and the KPI shape was linked to that cell.

**Outcome:**

The KPI became dynamic and responded to slicer selections.

**9. Analytical Work Performed**

The analysis was divided into three separate sections.

**9.1 Company Analysis**

Total company records: **263,295**

The company dataset was analyzed using the following fields:

- CIN
- Company Name
- Company Registration Date
- Company Category
- Company Class
- Listing Status
- Authorized Capital
- Paid-up Capital
- Company ROC

- Company Address
- PIN Code
- Company State
- Company Status
- Company Sub Category
- Company Industrial Classification

**Major analytical areas**

1. Company classification analysis
2. Listing-status analysis
3. Corporate-status distribution
4. Industry classification
5. Company registration trends
6. Authorized capital analysis
7. Paid-up capital analysis
8. Geographical distribution

**Company Class Findings**

| **Company Class** | **Records** |
|---|---|
| One Person Company | 4,563 |
| Private | 238,999 |
| Public | 19,733 |
| **Total** | **263,295** |

Private companies represented the largest category in the available company dataset.

**9.2 LLP Analysis**

Total LLP records: **33,932**

The LLP dataset was analyzed using:

- LLPIN
- LLP Name
- Registration Date
- Company ROC

- Company State
- Company District
- Company Status
- Company Industrial Classification
- Company Address

**Major analytical areas**

1. LLP registration trends
2. District-wise LLP distribution
3. Industry-wise LLP distribution
4. LLP corporate-status analysis
5. Geographical data-quality assessment

**LLP Registration Trend**

The available dataset showed the following yearly registration counts:

| Year | LLP Registrations |
|---|---|
| 2009 | 0 |
| 2010 | 14 |
| 2011 | 102 |
| 2012 | 178 |
| 2013 | 217 |
| 2014 | 385 |
| 2015 | 840 |
| 2016 | 1,780 |
| 2017 | 2,097 |
| 2018 | 2,388 |
| 2019 | 2,052 |
| 2020 | 2,167 |
| 2021 | 1,967 |
| 2022 | 2,789 |

| | |
|---|---|
| 2023 | 1,830 |
| 2024 | 2,696 |
| 2025 | 3,544 |
| 2026 | 4,753 |

The highest count in the available extract was recorded in 2026, with 4,753 LLP records.

This represents the available dataset extract, not necessarily a completed full-year registration count.

**LLP District Analysis**

The LLP dataset was also analyzed geographically using district-wise counts.

The analysis identified Kolkata as the largest recorded district category, with 21,595 LLP records.

Other district counts were examined to understand the distribution of LLP registrations across West Bengal.

Inconsistent names, duplicate geographical labels and null district values were identified as data-quality considerations.

Unverified district assignments were not treated as confirmed district-level results.

**9.3 Foreign Company Analysis**

Total source records: **5,385**

The Foreign Company Master dataset was analyzed separately using:

- FCRN
- Company Name
- Company Industrial Classification
- Company Country
- Company ROC
- Company Status
- Company Address

**Major analytical areas**

1. Country-wise distribution
2. Industry classification
3. Corporate status

4. Address-based geographical identification
5. Foreign-company presence in West Bengal

The dataset did not contain a registration-date field, limiting chronological registration analysis.

The India-wide geographical scope was retained when interpreting the 5,385 source records.

**10. Dashboard Development**

After completing the data preparation and analysis, an interactive Excel dashboard was developed.

The dashboard was designed to present important information in a consolidated visual format.

**Dashboard Components**

**1. KPI Cards**

Used to display important corporate metrics.

**2. Company Analysis Charts**

Used to represent company classifications, listing categories and corporate status.

**3. LLP Analysis Charts**

Used to visualize industry distribution, registration patterns and district-wise LLP counts.

**4. Foreign Company Charts**

Used to examine country-wise and status-wise distributions.

**5. Geographical Map**

A West Bengal district-level map was incorporated to represent LLP distribution geographically.

**6. Interactive Slicers**

Slicers were connected to relevant PivotTables and charts to enable dynamic filtering.

**7. Dynamic KPI Cards**

Helper PivotTables and GETPIVOTDATA were used to connect KPI values to slicer selections.

**11. Interactive Dashboard Functionality**

One of the important outcomes was making the dashboard interactive instead of presenting only static charts.

For example, when the Company Class slicer is used:

- Selecting One Person Company filters the connected analysis.
- Selecting Private filters the connected analysis.
- Selecting Public filters the connected analysis.
- Clearing the filter restores the overall result.

The helper PivotTable updates with the slicer selection, and the linked KPI displays the corresponding filtered total.

Separate slicer connections are maintained for the Company, LLP and Foreign datasets.

**12. Business Insights and Interpretation**

The project provides a structured way to understand the corporate landscape of West Bengal.

The available analysis supports several observations:

1. Private companies form the largest category in the Company Master dataset.
2. Kolkata represents the largest recorded LLP district category.
3. LLP registration counts vary considerably across the available years.
4. Industry classification helps identify the sectors represented in the LLP dataset.
5. Corporate-status analysis helps distinguish active, inactive, dissolved and other recorded statuses.
6. Capital analysis provides an additional perspective on the financial scale of registered companies.
7. Geographical standardization is essential for reliable district-level reporting.

These observations describe the available MCA records and should not automatically be interpreted as a complete measure of economic activity or current operational business activity.

**13. Data Limitations and Considerations**

Several limitations were considered while interpreting the results.

- Foreign Company Master data is India-wide and must be filtered by verified West Bengal addresses for state-level conclusions.
- Some district names have alternative spellings and require standardization.
- Records with missing or ambiguous district information cannot be assigned confidently without further verification.
- The 2026 registration figures represent the available data extract rather than a completed calendar year.

- Foreign-company registration trends cannot be calculated because the dataset does not provide a registration-date field.
- Company and LLP status fields represent the recorded MCA status and may not independently establish the present operational condition of every business.
- Capital values describe registered capital information and should not be interpreted as revenue, profit or market valuation.

**14. Final Project Outcome**

The project successfully established a structured Excel-based corporate analytics workbook using three MCA datasets.

The final deliverables include:

- Organized raw-data worksheets
- Cleaned datasets
- Data-quality checks
- Formula-based analytical worksheets
- Company analysis
- LLP analysis
- Foreign-company analysis
- Consolidated analytical summary
- PivotTables and PivotCharts
- Interactive dashboard
- Dynamic KPI functionality
- District-level geographical visualization

The project demonstrates how Microsoft Excel can be used to transform large, unstructured business datasets into an organized and interactive analytical reporting solution.

**15. Skills Demonstrated**

**Technical Skills**

- Microsoft Excel 2021
- Power Query
- Data Cleaning
- Data Transformation
- Data Validation

- Excel Formulas
- XLOOKUP and GETPIVOTDATA
- COUNTIFS and SUMIFS
- PivotTables
- PivotCharts
- Interactive Slicers
- KPI Development
- Dashboard Design
- Data Visualization
- Exploratory Data Analysis

**Analytical Skills**

- Data Interpretation
- Trend Analysis
- Classification Analysis
- Geographical Analysis
- Data Quality Assessment
- Business Insight Development
- Problem Solving
- Analytical Reporting

**16. Conclusion**

West Bengal Corporate Intelligence demonstrates an end-to-end practical application of Data Analytics using Microsoft Excel.

The project began with the collection of large corporate datasets from the MCA portal and progressed through data cleaning, transformation, quality assessment, analytical calculations, PivotTable development and interactive dashboard creation.

Throughout the process, practical challenges involving inconsistent dates, Power Query errors, PIN code formatting, duplicate records, geographical standardization and dynamic KPI functionality were addressed.

The final solution provides a structured way to explore company registrations, LLP distribution, industry classifications, corporate status, capital information and geographical patterns.

Beyond the final dashboard, the project provided practical experience in handling real-world datasets, investigating data-quality problems, building analytical summaries and communicating findings through visual reporting.

**The overall objective was achieved: transforming raw corporate registration data into a structured, interactive and insight-oriented analytical solution.**